

SECTION A /20 Marks**QUESTION 1.**

(a). Find the 23rd term of the arithmetic progression with first term 11 and the common difference -3 **(2 Marks)**

(b). Let $f(x) = \frac{1}{x^2 - 7}$ and $g(x) = \sqrt{x - 3}$. Find the domain of $f \circ g$. **(3Marks)**

(c). Find $\frac{dy}{dx}$ if $2y = x^2 + \sin y$ **(3Marks)**

(d) Find the area of the region enclosed by parabolas $y = x^2$ and $y = 2x - x^2$. **(4 Marks)**

(e) Find the volume of the solid formed when the area under $y = x^2$ between $x = 1$ and $x = 2$ is rotated about the x axis **(3 Marks)**

(f) Consider vectors $A(1,4,-7)$ $B(2,-1,4)$ and $C(0,-9,18)$. Evaluate $A \cdot (B \times C)$ **(3 Marks)**

(h). What are the values of a parameter λ such that the following matrix is a singular

matrix $\begin{pmatrix} 17 & 1 & -23 \\ \lambda & 0 & 6 \\ 0 & 0 & \lambda - 1 \end{pmatrix}$ **(2 Marks)**

SECTION B / 30 Marks**QUESTION 2.**

a) Find the interval and radius of convergence of the series **(5Marks)**

$$\sum_{n=1}^{\infty} \frac{(x-5)^n}{n^2}$$

(b) Show that the vectors $v_1(1,1,0)$, $v_2(1,0,1)$ and $v_3(3,1,1)$ form a basis in $\mathbb{R}^3$ (4 Marks)

c) Find the length of the curve $y = 1 + 6x^{\frac{3}{2}}$ from $x = 0$ to $x = 1$. (6 Marks)

QUESTION 3.

(a) Find where, if anywhere the function $f(x) = 3x^5 - 5x^3$ is increasing, decreasing, concave up, concave down ; inflection , local maximum and minimum points. 8MARKS

(b) Consider the following mapping $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined as $T(\vec{x}) = \langle x_2, x_1, x_1 + x_2 \rangle$.

Show that T is a linear operator? (4 Marks)

(c) Evaluate $\int_1^2 \frac{dx}{1-x}$ (3MARKS)

QUESTION 4.

(a) Find the eigenvalues and their corresponding eigenvectors of the matrix

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -2 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \quad (7 \text{ MARKS})$$

(b) Find the Taylor series generated by the function $f(x) = \frac{1}{x}$ at $x = 2$. (5 MARKS)

(c) Evaluate $\lim_{x \rightarrow 1} \left(\frac{\sqrt[3]{x}-1}{\sqrt[4]{x}-1} \right)$ (3 M3pARKS)